3. Sweep up material and place in proper DOT-approved container. Use more decontamination solution to clean remaining surfaces and also place this residue in container.
4. Loosely apply lid. Do not seal for 48 h, since gas generation may occur during neutralization. Isolate container in a well-ventilated place.
5. Discard all contaminated clothing. Decontaminate personnel and equipment using approved procedures.

Decontamination of empty containers:

1. Spray or pour 1–5 gal of decontamination solution into the container. Ensure that the walls are triple rinsed.
2. Leave container standing unsealed for a minimum of 48 h to allow for a complete neutralization of TDI.

Disposal:

1. Care should be taken to prevent environmental contamination from the use of this material.
2. Dispose of contaminated product, empty containers and materials used in cleaning up leaks, spills, or containers in a manner approved for this material.
3. The user of this material has the responsibility to dispose of unused materials, residues, and containers in compliance with all relevant federal, state, and local laws and regulations regarding treatment, storage, and disposal for hazardous and nonhazardous wastes.
4. Ensure that drums are labeled with correct hazardous waste code. Waste code U223.

Chillers; Crystallizers; Chemical Separation Method; Alternative to Distillation/Fractional Distillation*

Crystallization: An Alternative to Distillation

Many organic mixtures may be separated by cooling crystallization. In simple terms, cooling crystallization means that a mixture of organic chemicals is partially crystallized by reduction in temperature, without removal of any of the components by evaporation. An example of a crystallizer is illustrated in Fig. C-41.

Crystallization is a one-way process: the heat is removed, crystals are formed, and the mixture of crystals and solids are then separated. Many crystallizations take place at near ambient temperature so there is little heatup or cooldown required to get the right conditions for the separation to start.

Distillation, on the other hand, is a refluxing operation, where products are repeatedly evaporated and recondensed. Most distillations take place at elevated temperatures, which means that the materials being processed must be heated up and cooled back down again, usually with energy losses both ways. Also many distillations are run under vacuum to achieve better separation, which is energy intensive.

The latent heat of fusion in crystallization is generally much lower than the latent heat of vaporization. Since the latent heat must be removed only once, instead of many times as in distillation, the energy requirements are drastically lower for crystallization.

In the great majority of crystallizations, the crystals that form are 100 percent pure material, as opposed to something only slightly richer than the feed material

* Source: Armstrong Engineering Associates, USA. Adapted with permission.



FIG. C-41 Fatty chemical crystallizer with both brine and boiling refrigerant cooling. (Source: Armstrong Engineering Associates.)

as in distillation. With crystallization it is not necessary to repeatedly melt and refreeze to obtain high purity. The pure crystals may have some impure mother liquor on the surfaces and sometimes contained within the crystals as occlusions. However, the purity increase is extremely rapid and normally one or perhaps two crystallizations can give very high purities.

In addition to much lower energy costs as compared to distillation, crystallization has other significant benefits, such as:

- Low-temperature operation, which means low corrosion rates, and often the use of less costly alloys compared to evaporation-based separations. The low-temperature operation also means little or no product degradation, which for heat-sensitive materials may be crucial. There is no formation of tars, which represent a yield loss, a severe waste disposal problem, and usually requires additional separation equipment and energy for the tar removal in order to give the desired product color.
- Enclosed systems with little or no chance of leakage of dangerous or noxious fluids. The systems are normally simple and require few pieces of equipment and little instrumentation.
- Favorable equilibrium; often the freezing points of organic chemicals are widely spread enabling easy separation by crystallization, where separation by distillation may be extremely difficult.
- High purity; the crystals that form in a great majority of cases are 100 percent pure material. While impurities may adhere to crystal surfaces, or be included inside the crystal, recrystallization usually produces very high purities with relative



FIG. C-42 Stainless steel process side crystallizer for fatty chemicals—shell side has stainless steel for corrosive coolant in lower section. (Source: Armstrong Engineering Associates.)

ease. The normal product purity range is 95 to 99.5 percent, although higher figures are often reached. One large plant produces 99.9+ percent pure product.

- The scraped surface crystallizer makes crystallization continuous. Generally, the only reason to work with batch crystallization is very low design capacity. If design capacity is above 500,000 lb annually, the scraped surface continuous crystallizer will save time, energy, and manpower.

Many crystallizations are performed using batch cooling in stainless steel or glass-lined kettles (Fig. C-42). By and large this represents continued growth from specialty chemical to commodity, with little engineering attention paid to the crystallization part of the process.

This method offers significant advantages over batch crystallization, such as:

- Smaller equipment, which generally means less expensive installations, less floor space needed, less operator labor, and no duplication of instrumentation, piping, etc.
- Better process control, less upsets of hazardous or expensive materials, and less peak utility demand

Many continuous crystallizations are done in evaporative crystallizers based on designs typically used for inorganic chemicals. With inorganics there is usually a very flat solubility curve, which means that a change in mixture temperature produces relatively few crystals. Other continuous crystallizations are sometimes performed by cooling and partially crystallizing in shell and tube exchangers, which can foul, requiring them to be taken out of service for cleaning.



FIG. C-43 Crystallizer for very viscous medium with individual drive gear motors. (Source: Armstrong Engineering Associates.)

The scraped surface continuous crystallizers offer many advantages over these other methods of continuous crystallization, such as:

- Modular design allows for easy expansion with growth in demand.
- Simple, self-contained construction with minimum instrumentation and auxiliaries, such as: condensers, vacuum systems, etc.
- May be run for extended periods between hot washings where many shell and tube exchangers would plug up in minutes.
- May be run at much higher temperature differences between process fluid and coolant than could ever be attempted with shell and tube equipment without serious fouling or plugging.
- May be used over an extremely wide temperature range, from –75 to +100°C. It is usually very difficult to run vacuum crystallization equipment over a broad range of temperatures.
- May be used with high percent solids. Vacuum crystallizers are normally limited to about 25 percent by weight or less solids. This equipment has worked in a range of 65 percent by weight solids as slurry.
- High viscosities are not a problem, with several crystallizations being carried out from mother liquor with viscosities of 10,000 cp or higher (see Fig. C-43).



FIG. C-44 Crystallizer for separation of aromatic isomers. (Source: Armstrong Engineering Associates.)

- Flow pattern in once-through operation closely resembles plug flow so conversion of batch to continuous processes is easy, and virtually any desired time/temperature pattern is possible.
- In small-capacity cases, a scraped surface crystallizer will be very inexpensive. This is also true in cases where, for much larger installations, vacuum crystallization may seem most attractive.

Wide range of capacities

Scraped surface crystallizers have been used over a wide range of capacities, from the smallest continuous operations (typically about 1000 tons/year) up to 250,000 tons/year. There is no practical upper limit to capacity.

Good solubility curve

Cooling crystallizations are obviously most advantageous where the solubility curve will produce good yields with simple cooling of the mixture. This is true of a wide variety of organic mixtures. See Figs. C-44 and C-45.

Low-temperature crystallizations

Scraped surface continuous crystallizers offer the best approach for low-temperature crystallizations such as: the separation of meta- and paraxylenes or oleic and linoleic acids.



FIG. C-45 Stainless steel process side crystallizer for oligomers formed in fiber processing—three separate process duties are included. (Source: Armstrong Engineering Associates.)

Products with high boiling point rise

Some mixtures of inorganic chemicals in water show very high boiling point rises as concentration proceeds, reducing the vapor pressure, and dramatically increasing the vacuum requirements. Many such mixtures produce abundant crystal growth on cooling. Often a scraped surface continuous unit may be used in conjunction with a vacuum unit, with the vacuum unit doing the high-temperature part of the crystallization and the scraped surface unit doing the low-temperature part.

Products with similar vapor pressures

Many aromatic chemicals, particularly isomers, have nearly identical vapor pressure characteristics, which makes distillation very difficult. However, these same mixtures often have widely varying freezing points, which makes crystallization simple and effective.

High viscosity fluids

High viscosity, due either to high mother liquor viscosity or high percent solids, does not present problems to the scraped surface continuous crystallizer but may make other types of crystallizers totally inoperable.

Severe fouling

The fouling tendencies of many slurries are overcome because the deposits on the heat transfer surfaces are continuously removed.

The following list of compounds is incomplete because in some cases manufacturers are not made aware of the material they are working with, and in other cases, manufacturers are bound by secrecy agreement not to discuss the use of equipment with a specific product.

- Anthracene
- Anthraquinone
- Benzene Hexachloride
- Benzoic Acid
- Bisphenol A
- Butyl Cresol
- Butyric Acid
- Caffeine
- Calcium Nitrate
- Caprolactam
- Cyanoacetamide
- Dibutyl Cresol
- Diglycerides
- Dimethyl Hydantoin
- Dimethyl Terephthalate
- Fatty Acids
- Lactose
- Laurolactam
- Levulinic Acid
- Menthol
- Methionine
- Monoglycerides
- Naphthalene
- Nitrochlorobenzene
- Oligomers
- Palm/Palm Kernel Fats
- Paracresol
- Paradichlorobenzene
- Paraxylene
- Pentaerythritol
- Potassium Chloride
- Potassium Nitrate
- Sebacic Acid
- Silver Nitrate
- Sodium Carbonate
- Sodium Lauryl Sulfate
- Sodium Sulfate
- Sorbic Acid
- Sterols
- Tall Oil Fatty Acids
- Tallow Fatty Acids
- Tetrachlorobenzene
- Tetramethyl Benzene
- Vitamins
- Waxes

As mentioned earlier, there are substantial differences between processes. Crystallizers are designed to handle a specific duty. What might be right for one application may not be appropriate for another.

The following are examples of applications that require different approaches to achieve the separation of materials by cooling crystallization.

Separation of chlorobenzenes

Para- and orthodichlorobenzene, which are used in the example on solubility thermodynamics, represent two important chemical products that lend themselves to separation by cooling crystallization. The paraisomer crystallizes at temperatures far above the point where either ortho crystals, or the eutectic is reached. Paradichlorobenzene forms extremely tough crystals, which adhere readily to any cooled surface, requiring vigorous scraping to remove them. These tough crystals can stand a certain amount of abuse without degradation in size.

Normally the mixture produces a very thick slurry. Great care must be exercised to handle it. The extremely steep solubility curve presents many opportunities for good crystal growth. However, there is a danger of uncontrolled crystallization, which must be handled carefully or the entire unit may freeze solid.

Strong equipment, and ingenious slurry handling, often with staged operations, are the basics of this process and similar separations of xylene isomers, cresols, and other disubstituted benzenes. (See Fig. C-46.)

Separation of fatty materials

Fatty acids from tallow or tall oil, mono-, di-, and triglycerides, fatty alcohols, and related compounds all may be separated by crystallization when other separation

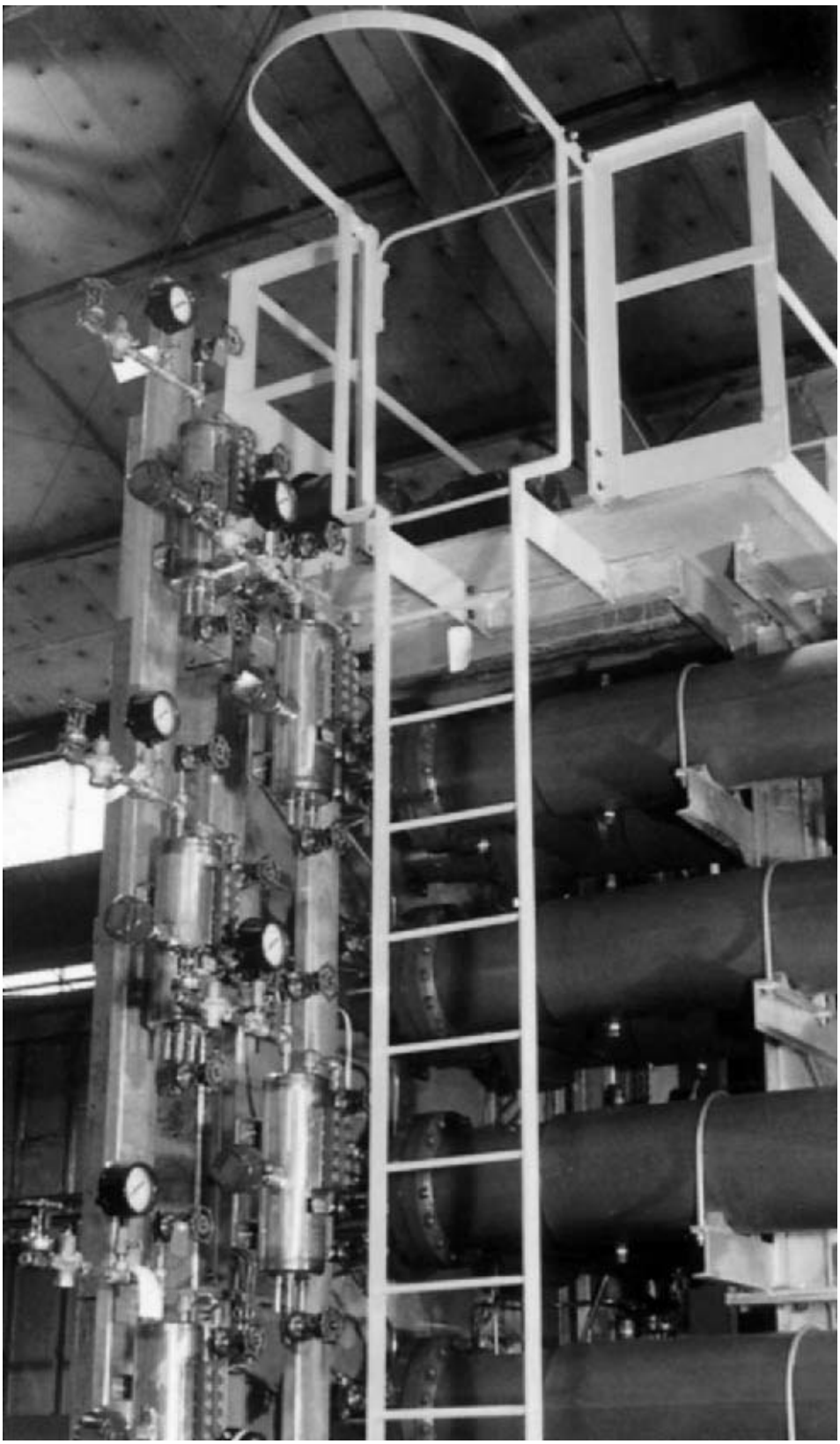


FIG. C-46 Drive end of a special unit which includes mechanical seal systems. (Source: Armstrong Engineering Associates.)



FIG. C-47 Large installation of wax crystallizers in a petroleum refinery. (Source: Armstrong Engineering Associates.)

methods will not work. The extremely delicate nature of the crystal and the sensitivity to shear, which can rapidly produce an inseparable crystal, must be taken into account when separating these materials.

The time/temperature relationship is also of extreme importance, sometimes requiring sophisticated cooling arrangements on the shell sides of the equipment.

Solvents are sometimes used to obtain optimal separations, although solvent-free separations using detergents to separate saturated and unsaturated compounds have also been frequently used.

With this process, crystal growth is relatively slow. Care must be exercised to allow time to grow a decent crystal, which may be easily separated. Reducing shear is more important than producing a rugged machine for handling these delicate materials.

Dewaxing lubricating oil represents the largest use of scraped surface continuous crystallizers (Fig. C-47). Wax has the same boiling point range as lubricating oil fractions, but has a much higher freezing point. Therefore, cooling crystallization is a very effective way to separate the two materials.

Many of the processing plants are quite large and require many scraped surface continuous crystallizers, often with a number of units in a series. Larger plants usually require several parallel trains of crystallizers.

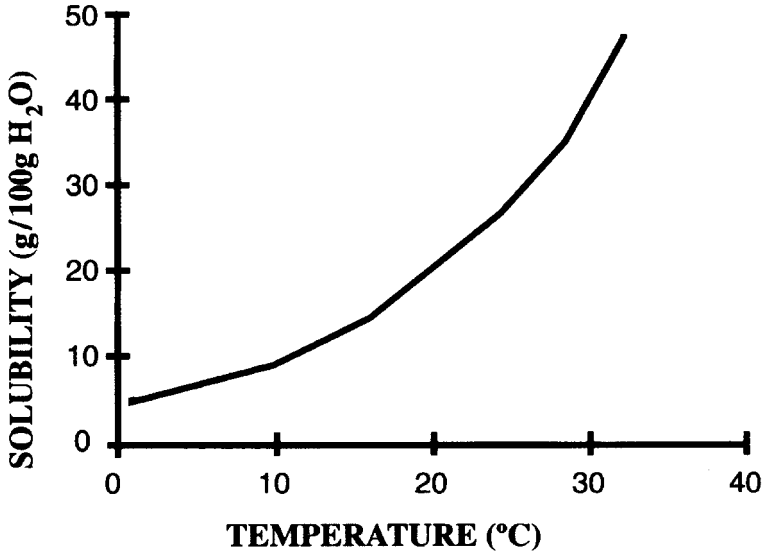


FIG. C-48 Solubility of Na_2SO_4 in H_2O. (Source: Armstrong Engineering Associates.)

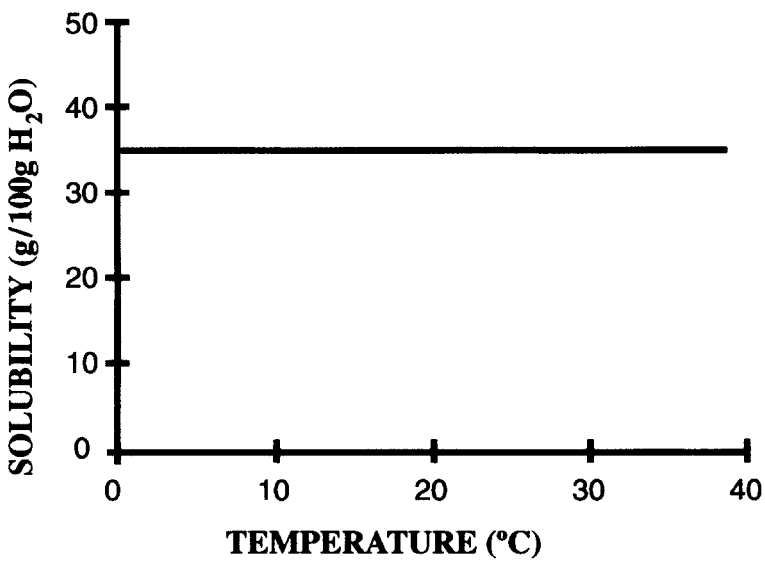


FIG. C-49 Solubility of NaCl in H_2O. (Source: Armstrong Engineering Associates.)

The basic goal of designing scraped surface continuous crystallizers for dewaxing is to ensure longer time on stream between turnarounds.

Some inorganic chemicals have a steep solubility curve with temperature, i.e., a small amount of cooling produces a substantial crystal yield. Such materials are well suited for cooling crystallizations. A typical such solubility curve is shown in Fig. C-48.

Many inorganic compounds have relatively flat solubility curves as shown in Fig. C-49. These compounds are not well suited to cooling crystallization. Vacuum crystallization is the best method of separation for these mixtures.

However, there are some important cases with good characteristics for continuous cooling operations, using scraped surface crystallizers. Some examples of where scraped surface continuous crystallizers offer advantages include: sodium sulfate, potassium nitrate, sodium carbonate, nickel sulfate, ammonium thiosulfate, calcium nitrate, as well as many other inorganic compounds.

Many such processes have been relatively small scale, however some extremely large facilities have also been built. There is no practical upper limit to equipment capacity. The starting cost is modest, and expansion on an incremental basis is simple and often attractive.

The method of cooling can be either direct jacket side boiling refrigerant or brine cooling, depending on the temperature requirements.

Solubility Thermodynamics

In order for cooling crystallization to be an attractive method of separation, it is necessary that one component come out of a solution as the temperature changes. This can be determined by solubility thermodynamics. Understanding these relationships is fundamental to the equipment design.

The ideal case for crystallization

There are a number of frequently encountered cases where the ideal liquid mixture assumptions are applicable. In such cases the solubility, and therefore the ease of separation, can be easily calculated. Many of these cases are reaction mixtures that do not lend themselves to conventional methods of separation. Some frequently encountered examples are:

- Mixed xylenes
- Mixed chlorobenzenes
- Paraffins
- Many multisubstituted benzenes

The nonideal case for crystallization

There are a number of cases where the ideal liquid mixture assumptions are not true.

These include:

- Polar solutes in polar solvents, such as fatty acids in acetone
- Polarized solutes in polar solvents, such as naphthalene in methanol
- Dimerization or hydrogen bonding, such as many organic acids

Prediction of solubility in the ideal case

Under those conditions where the ideal liquid mixture assumptions can be considered to hold, the solubility relationship is quite simple. In the ideal case, the solubility curves and eutectics can be fairly accurately predicted using the Van T'Hoff equation:

$$L_n X_a = \frac{\lambda_a}{R}\left(\frac{1}{T_o} - \frac{1}{T}\right)$$

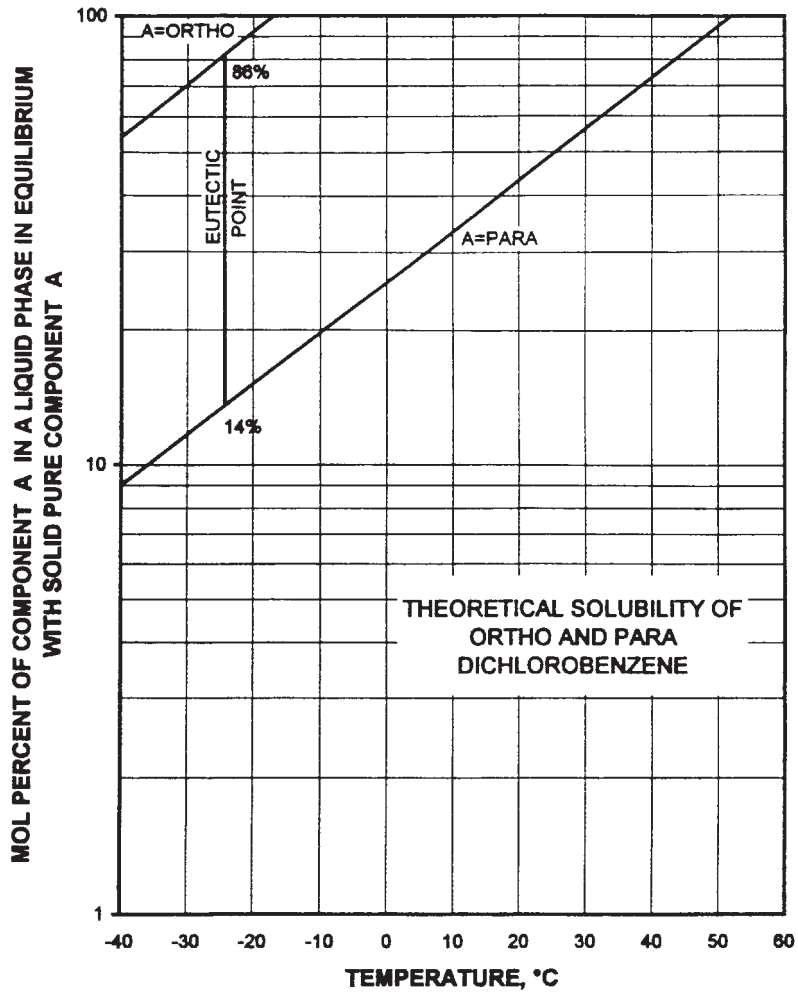


FIG. C-50 Theoretical solubility of ortho- and paradichlorobenzene. (Source: Armstrong Engineering Associates.)

where

X_a is the mole fraction in solution of component a
λ_a is the molar heat of fusion of component a
R is the gas constant
T_o is the melting temperature of component a at the system pressure
T is the system temperature

Therefore given the melting point of a substance and its molar heat of fusion, it is possible to predict its solubility in an ideal mixture and, by judicious use of these results, predict the eutectic temperature and composition.

Numerical example

Figure C-50 illustrates a direct plot of the Van T'Hoff equation, relating the mole fraction of both ortho- and paradichlorobenzene in solution of an ideal mixture at the temperatures shown. This means that in the case of an ideal mixture of para in a solvent, the composition of the saturated liquid phase is as indicated by the

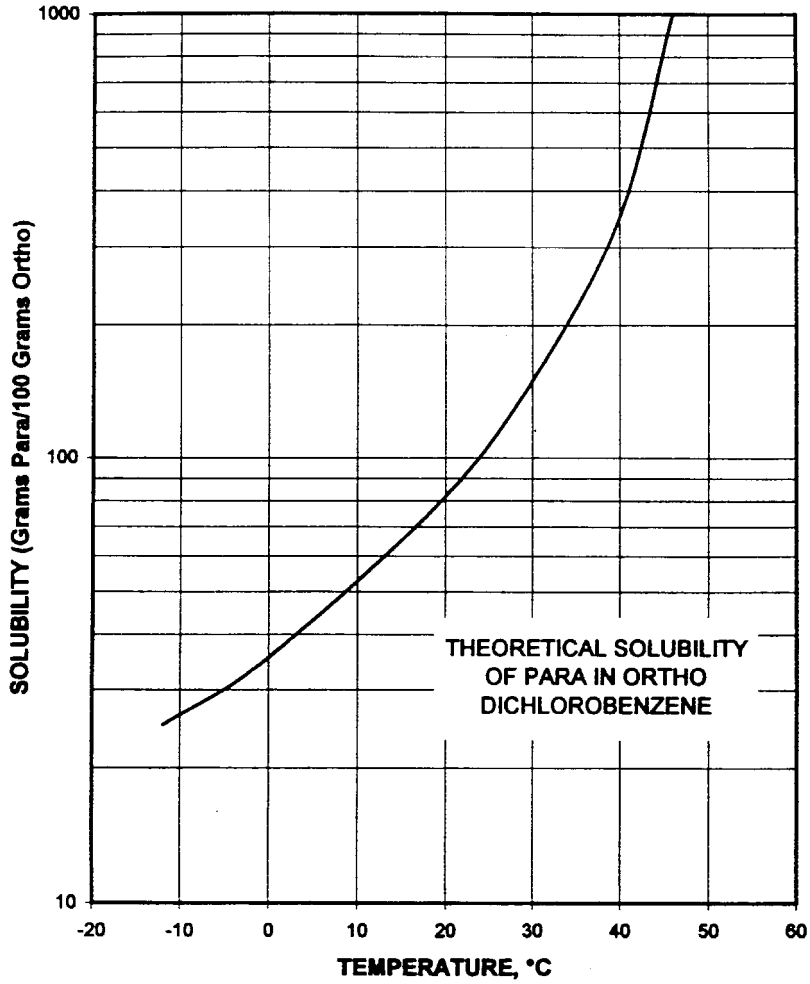


FIG. C-51 Theoretical solubility of para in orthodichlorobenzene. (Source: Armstrong Engineering Associates.)

para curve. For instance, this ideal liquid mixture would contain about 26 percent para at 0°C regardless of solvent, as long as the system is ideal. The ortho curve has the same significance.

A mixture of ortho and paraisomers is a frequently encountered case, and closely approximates the ideal predictions. In this case, mole and weight percents are identical because the two components are isomers. In this binary case, the solubility of para is read from its curve, and the ortho is the balance needed to sum up to 100 percent. The balance, which is the ortho percent, must be less than or equal to the percents indicated by the ortho curve, otherwise an impossible case of more than 100 percent could arise.

At the point where the sum of the ortho and para curves equal 100 percent, the eutectic occurs. Further change of composition for that system is impossible. There is a unique temperature where this will occur. At this point the system shows equilibrium between the liquid phase mixture of the two isomers, and solid pure para and also pure ortho as solid. The phase rule indicates this is a determinant point, and if the temperature is further lowered, solidification will take place,

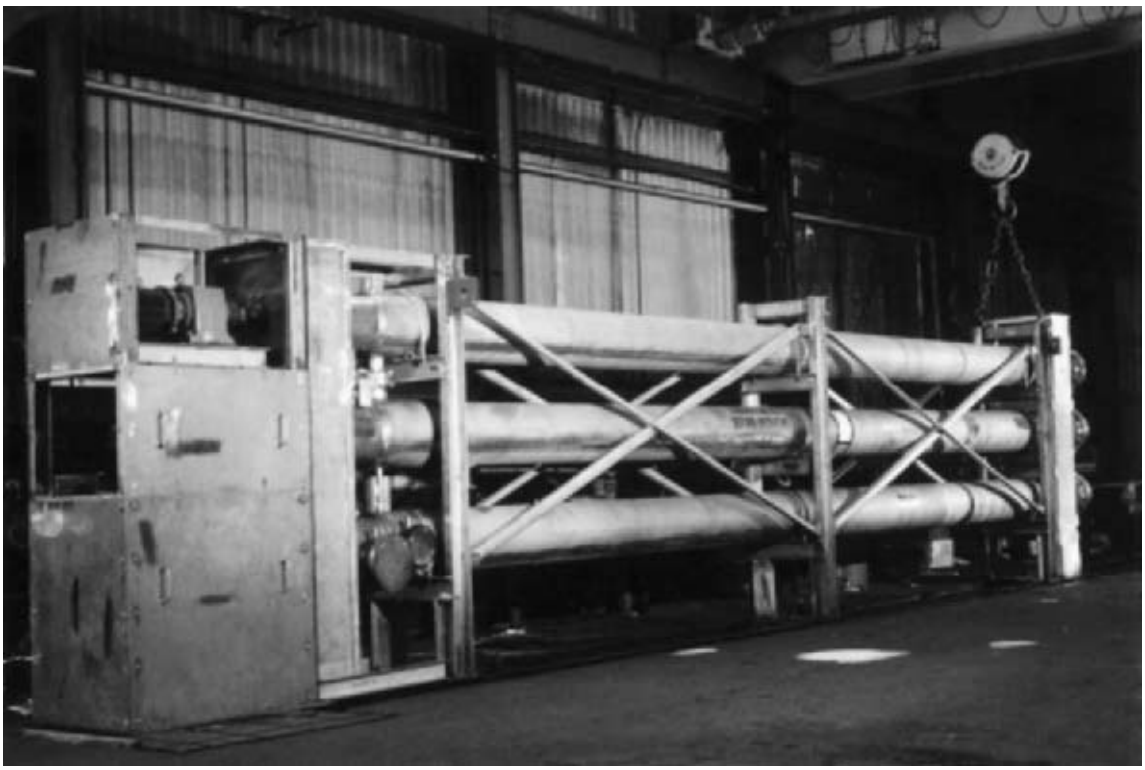


FIG. C-52 Special crystallizer made of all stainless steel, including structures and drive cabinet. (Source: Armstrong Engineering Associates.)

producing the homogeneous eutectic. As nearly as the plot can be read, this turns out to be at –24°C, with a composition of 14 percent para and 87 percent ortho. The International Critical Tables show the eutectic to be at –23.4°C, with a composition of 86.7 percent ortho, and 13.3 percent para, which is certainly in excellent agreement with published data.

Plotting the figure in another way, solubility expressed as grams para per 100 grams ortho will give the results most often used to calculate the recoveries, as illustrated in Fig. C-51.

Here it is clear that the best way to separate a mixture of these two components is to cool the mixture, driving the para out of the solution; as the cooling approaches the eutectic point nearly total recovery is attained.

A simple mass balance will show that a mixture of 65 percent para, 35 percent ortho can be reduced to the eutectic point with a theoretical recovery of around 90 percent of the para. At that point the mixture, ortho rich mother liquor can be sold, or perhaps the eutectic can be split, using solvents to produce pure substances.

While the calculations shown are theoretical in nature, the results are almost identical to published solubility data.

These calculations can be quite useful for basic feasibility work to evaluate whether crystallization is a viable alternative to other separation methods.

If this is the case, verification of the theoretical predictions is usually advisable. Verification is easy to carry out using simple laboratory equipment.

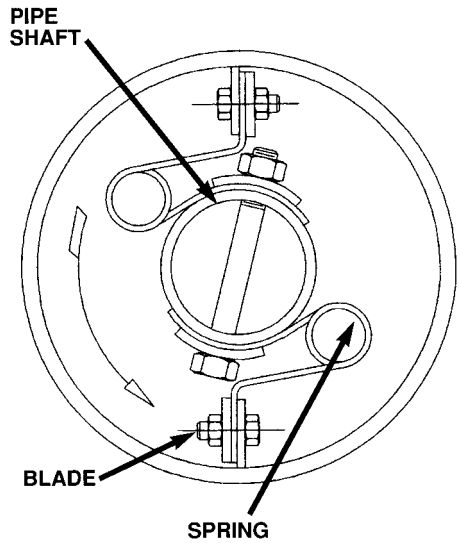


FIG. C-53 Internal arrangement in crystallizer. (Source: Armstrong Engineering Associates.)

Scraped Surface Continuous Crystallizer Specifically Designed for the Growth of Easily Separable Crystals

A scraped surface continuous crystallizer is built as a double pipe element, generally with a 6-, 8-, 10-, or 12-in diameter inner pipe and a larger diameter outer pipe. The area between the inner pipe and outer pipe is filled with cooling fluid. The inner pipe contains a rotating scraper blade element, which mixes the process fluid flowing through the inner pipe and removes deposits that form on the inner pipe wall as cooling occurs. (See special crystallizer in Fig. C-52 and internal arrangement in Fig. C-53.)

In effect, the scraped surface continuous crystallizer is a heat exchanger, but quite an unusual one, because it generally performs as a cooling crystallizer. Heat transfer occurs across the wall in the inner pipe, with cold fluid outside, and process fluid inside. As cooling occurs, crystals tend to form and foul the inner pipe wall. The scraper blades rotate on the inner pipe wall and remove the deposits that would inhibit heat transfer. The majority of the crystallization takes place in the bulk of the fluid, as opposed to the wall, thus allowing growth of easily separable crystals.

Generally, the scraper blades rotate at moderate speeds of 15 to 30 rpm. This speed promotes adequate heat transfer, consistent with good crystal growth and easy separation. A scraped surface crystallizer is sometimes confused with a margarine or ice cream machine since the principle is similar. The final result, however, is very different. Margarine and ice cream machines are also designed to crystallize materials, but the scraper blades rotate at much higher speeds, which produce higher heat transfer and tiny, inseparable crystals. This method is used to produce a homogenized product, but not for separating crystals.

Cooling is carried out by whatever coolant is selected and run through the annulus between the inner and outer pipe. Coolants may be cooling tower water, chilled brines, evaporating refrigerants, mother liquor stripped of crystals, or a combination of these.

A typical commercial-sized installation will consist of several double pipe elements, each with a length of 20 to 40 feet, connected in series. This provides a

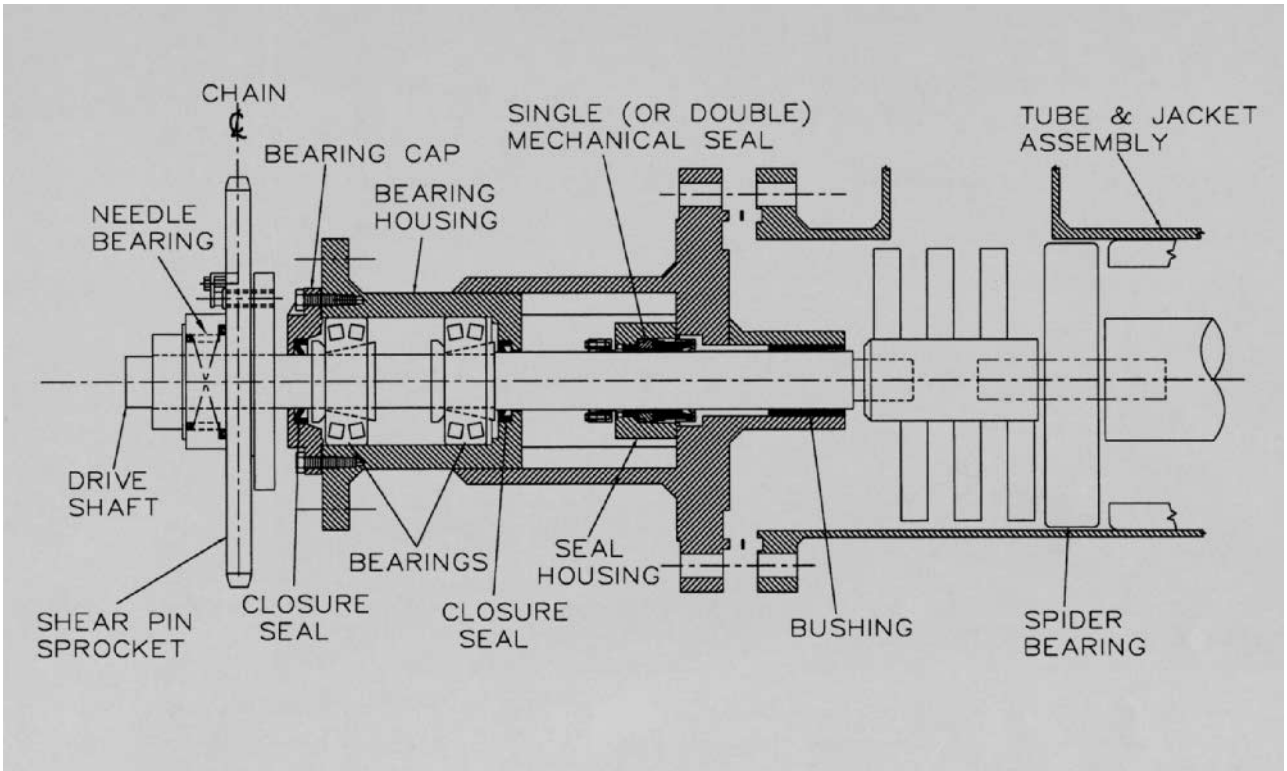


FIG. C-54 Bearing and seal assembly (crystallizer). (Source: Armstrong Engineering Associates.)

long thin flow path that promotes a close approach to plug flow, which is very important in many crystallizations.

Drive system

The drive system on a scraped surface continuous crystallizer must allow free turning of the driven shaft. The shaft must also be sealed against leakage of the process fluids. The drive system used is illustrated in Fig. C-54.

The drive housing is flanged and bolted to the inner pipe. Frequently, mechanical seals are used to avoid leakage around the driven shaft. It is imperative that the drive shaft is completely stabilized to avoid any bending, which would shorten the life of the seal. The use of two heavy-duty self-aligning bearings locks the drive shaft into place, and prevents any unwanted movement, which could cause leakage. This system has proven to be completely reliable in numerous installations around the world.

In concept this arrangement is quite similar to the drive system in many pumps, so maintenance personnel should be familiar with the system.

The use of square end matching parts between the drive shaft and the internal parts allows removal of the scraping element without any need to disturb the drive system. Conversely, the drive system can be maintained without disturbing the scraping mechanism.

Chimneys (*see* Stacks)

Cleaning

If chemical compounds need to be removed, many potential acidic, alkaline, or organic cleaners are available on the market. Environmental regulations should be carefully heeded, as there have been many recent changes with respect to carcinogenic content. These changes can be stated in both federal and state or provincial laws. The strictest of the different laws should be selected for compliance to minimize the risk of lawsuits and other potential problems.

Cleaning a surface might require just a water wash, a hot-water wash, or a steam jet. Temperature and purity of the water or steam used will depend on the material to be cleaned and the process. Cleaning may require soap or detergent of some kind.

Manufacturers' precautions need to be observed for reasons of safety and environmental considerations. For instance, many detergent companies now advertise their phosphate-free content to people supporting the environment. Although domestic detergents are not classified as poisonous, many large laundry facilities, such as those in large hotels, ask that clients conserve the use of linens and soap to reduce the effects of detergent solution in sewer effluent. Phosphates in detergents are among the "nutrients" in ocean environments, higher levels of which are attributed to accelerating the growth of "red tide" algae (given the right sunlight, salinity, and water temperature). Plankton species involved in "red tide" include *Ptychodiscus brevis*, which produces a poison called brevetoxin. In 1989, NOAA (National Oceanic and Atmospheric Administration, USA) stated that this poison had caused the death of dolphins on an unprecedented scale or sufficient weakening of their immune systems that they died from other less serious causes. This has consequences with respect to human health as dolphins are mammals very high on the food chain. The only mammals higher than dolphins are humans.

For additional material on air cleaning, *see* Air Filtration; Air Pollution Control; Air Purification.

For gas-turbine cleaning, *see* Turbines.

Specialized cleaning processes, and the corresponding customized equipment, are frequently required in industries such as agriculture and pharmaceuticals. Consult manufacturers of items, such as centrifuges and filters, to discuss specific options. Also consult Some Commonly Used Specifications, Codes, Standards, and Texts for cleanliness requirements for different applications.

Clutches (*see* Power Transmission)

Coatings (*see* Ceramics; Metallurgy)

Cogeneration

The term *cogeneration* means recovery of waste heat. Frequently, the term is used to describe modifications of turbine cycles and some form of recovery of waste heat from the exhaust gases of combustion.

The exhaust gases may be circulated through a heat-recovery steam generator (HRSG) and the resultant steam used to power a steam turbine. This is called a *combined cycle* application (gas turbine + HRSG + steam turbine). *See also* Turbines.

The exhaust gases may be led directly into a greenhouse to provide heat to growing flora. This is termed a *waste heat recovery* (WHR) cycle.

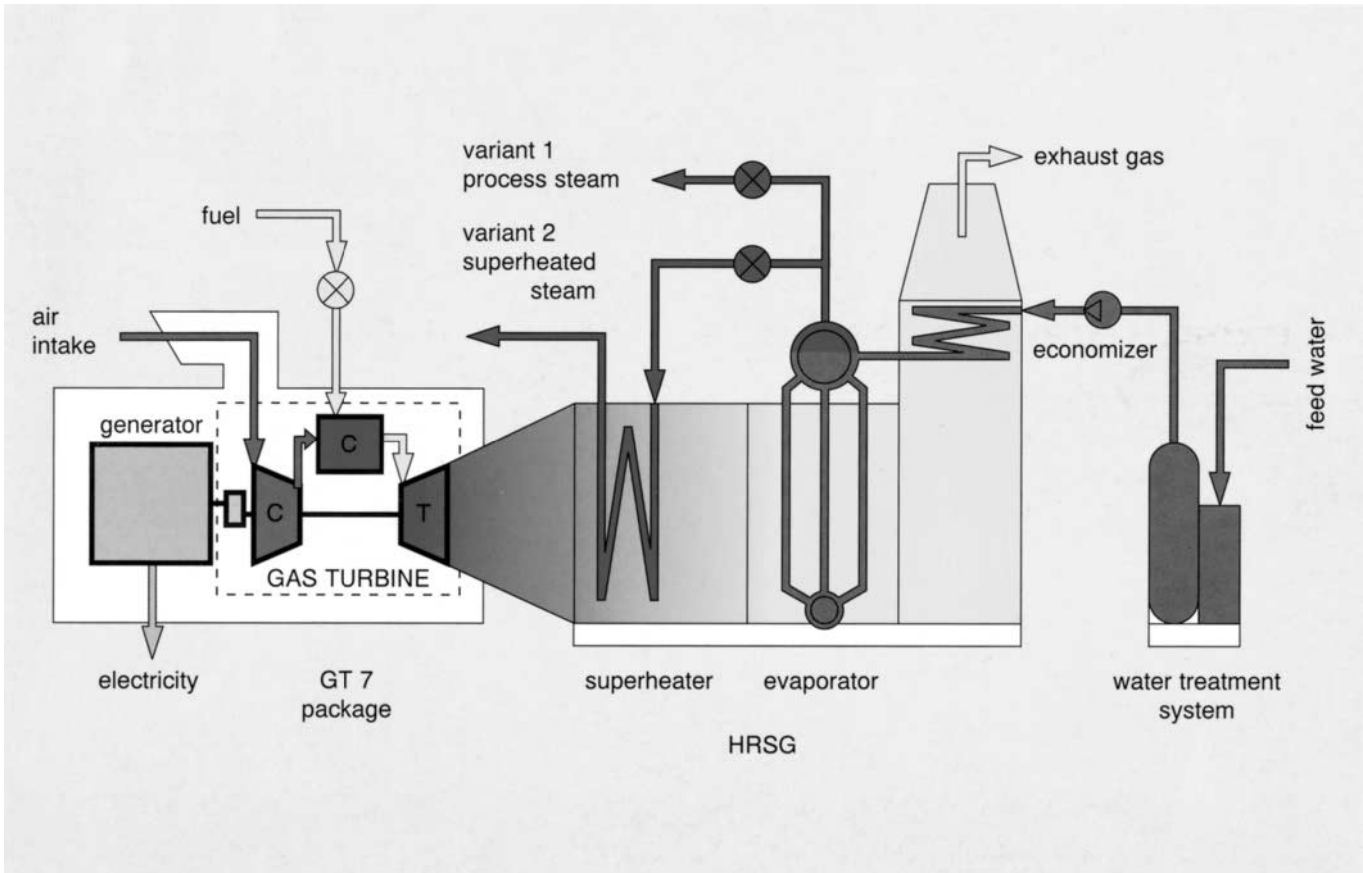


FIG. C-55 Cogeneration based on a GT 7 gas turbine. (Source: Alstom.)

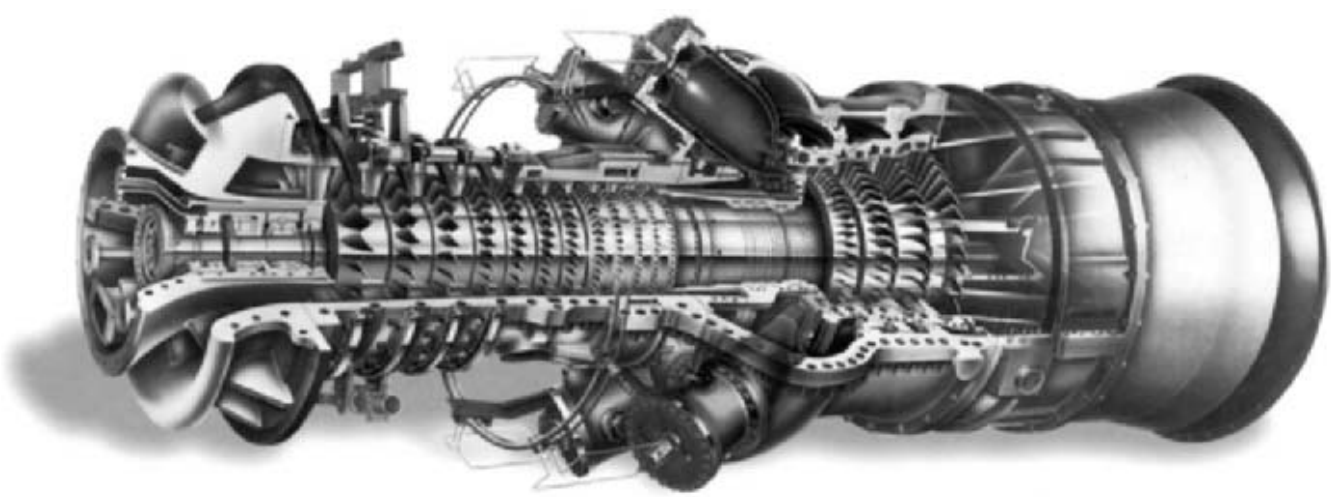


FIG. C-56 The GT 7 gas turbine. (Source: Alstom.)

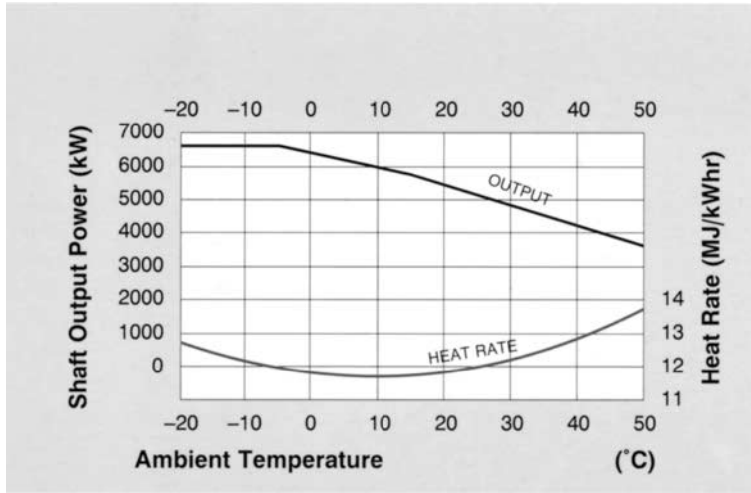
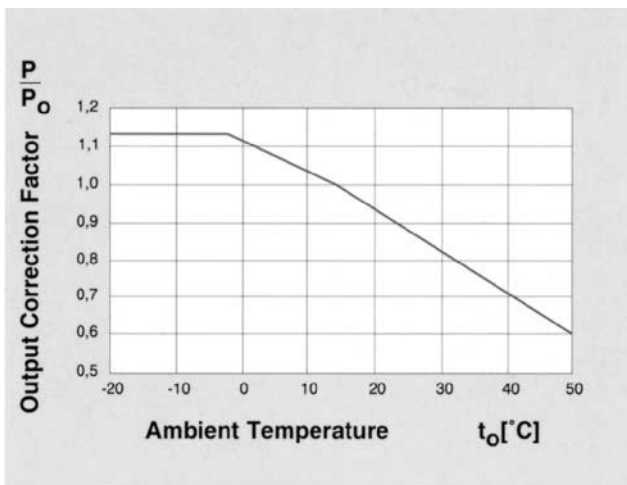
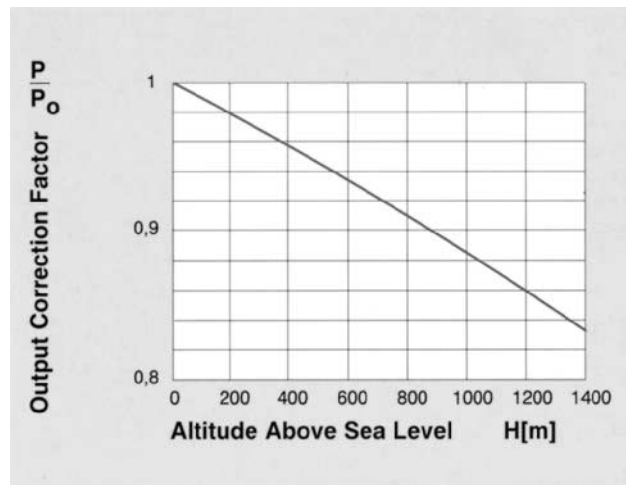


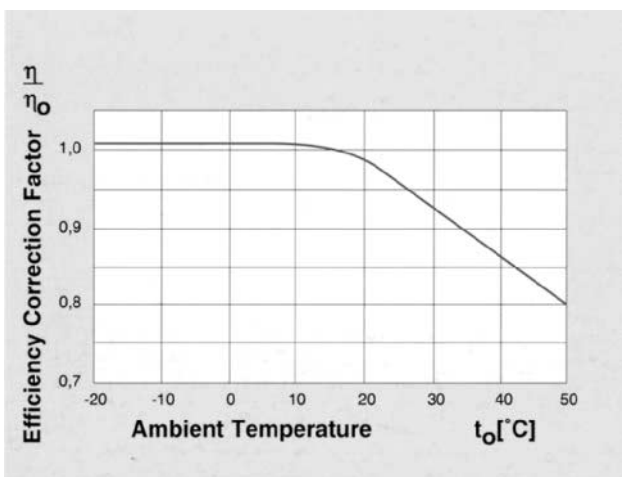
FIG. C-57 Nominal performance curve at gearbox output shaft for a GT 7. (Source: Alstom.)



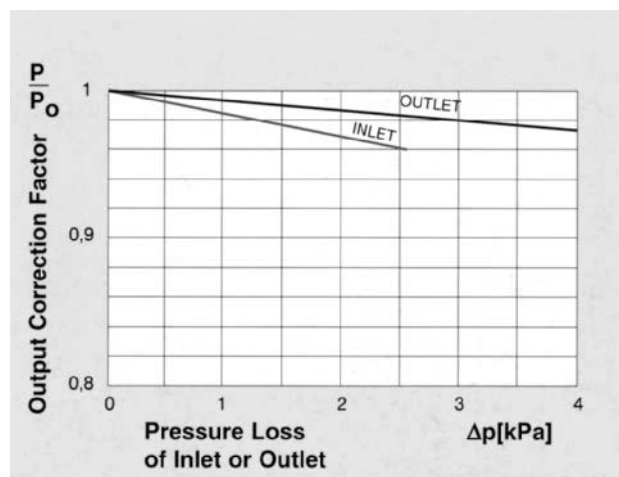
(a)



(b)



(c)



(d)

FIG. C-58 GT 7 gas turbine correction curves at gearbox output shift. (Source: Alstom.)

The exhaust gases may be circulated through a heat exchanger and the resultant heat used to provide household or industrial heating needs. This is a *heat and power* (H&P) application.

Figures C-55 through C-61 illustrate a typical H&P application using an Alstom GT 7 turbine. (*Note:* Alstom's GT 7 cogeneration plant is based on Kawasaki Heavy Industries gas turbine type M7A-01.)

The cogeneration unit can operate on either natural gas or liquid fuel with more efficient fuel utilization than separate generation of heat and power. The gas turbine exhaust is coupled to a heat recovery steam generator and turbine exhaust waste heat is used for heating purposes or process steam production.

The high turbine exhaust gas temperature—around 550°C—enables production of steam at high temperatures. Such steam may be required as process steam in the gas and oil industry and paper and pulp mills, as well as in the textile and food stuff industries.

The cogeneration unit based on a gas turbine is environmentally friendly—emissions in exhaust gas from the HRSG's stack meet current environmental regulations.

Modular configuration of the gas turboset, heat recovery steam generator, and accessories enables a quick assembly of the whole cogeneration plant at the site. The unit is designed for continuous base load service and suitable material ensures the long life cycle of hot gas path components.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.

Coker (*see* Oil Sands)

Color Coding

Color coding has been argued about among plant operators. It may occur in some plants, while others prefer not to use it. Basically, pipes are coded with a colored band and text on the colored band, according to "fire protection" (white letters on red), "dangerous" (black letters on yellow), "safe" (black letters on green), and "protective" (white letters on blue) potential. The width of the band will vary according to pipe size. Colored arrows indicate flow direction. See the safety specifications included in various standards (*see* Some Commonly Used Specifications, Codes, Standards, and Texts).

Columns (*see* Towers and Columns)

Combustor(s); Low NO_x Combustor (*see* Turbines)

Compressors

The two main subdivisions for compressors are turbocompressors (rotating) and reciprocating compressors. These two types fulfill different purposes in process plants. A rough rule of thumb is that *reciprocating compressors* are used in high-pressure ratio (pressure ratio = discharge pressure/suction pressure), low-volume applications. *Centrifugal compressors* were thought of as being high-volume, low-pressure ratio applications. In the 1970s, the commissioning of the Cooper RBB series barrel compressor design at the south Kaybob plant in northern Alberta then meant centrifugals were also suitable for high-volume, moderately high-pressure applications. Those particular compressors were used in

GT 7 Package for Cogeneration Application

Gas Turbine Engine Kawasaki M7A–01

Gearbox
Type: parallel
Output Speed: 1,500 rpm or 1,800 rpm

Generator
Cooling: air cooled
Power Factor: 0.8
Frequency: 50/60 Hz
Capacity: natural gas – 7 059 kVA
liquid fuel – 6 800 kVA
Insulation Class: F

Starter: hydraulic starting system

Gas Turbine Air System

Air Intake Unit
Filter
Silencer

Enclosure Ventilation
Ventilation Fan
Silencer
Filter

Enclosure
Type: outdoor type pre-fabricated
Steel Sound Attenuating Enclosure
Noise Level: 80 dB(A) at 1 m

Baseframe
Welded Frame: common to turbine, gearbox, generator and auxiliaries

Electrical/Control Panels
Gas Turbine Control Panel
Generator Panel
Synchronizing and Protection Panel
MV Measuring Panel
Generator Terminal Box
DC System

GT 7 Cogeneration Power Plant Performance

(Parameters valid for altitude 0 m, inlet duct loss 1.2 kPa, exhaust duct loss 3.0 kPa. Feed water temperature 60 °C, saturated steam pressure 1.8 MPa.)

Natural Gas

Inlet Air Temperature °C	Output at Generator Teminals kW	Fuel Consumption kW	Exhaust Gas Flow kg/s	Gas Turbine Exhaust Gas Temperature °C	Saturated Steam Production t/hr	Electrical Efficiency %
0	6,130	20,275	22.61	550	15.0	30.22
15	5,577	18,962	21.58	560	14.8	29.47
30	4,652	16,904	19.52	581	14.4	27.58
40	4,023	15,522	18.04	598	14.1	25.92

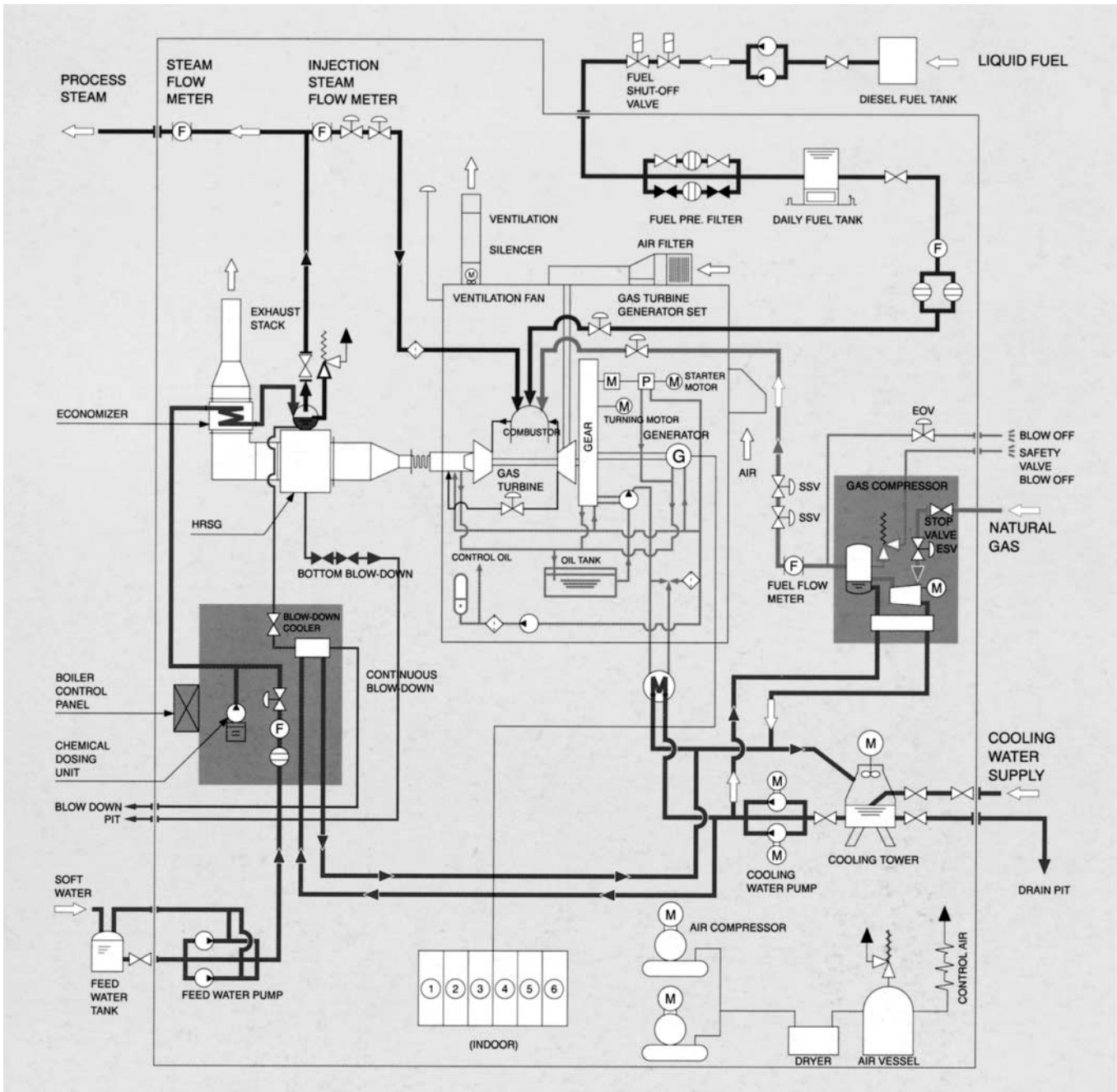
Liquid Fuel

Inlet Air Temperature °C	Output at Generator Teminals kW	Fuel Consumption kW	Exhaust Gas Flow kg/s	Gas Turbine Exhaust Gas Temperature °C	Saturated Steam Production t/hr	Electrical Efficiency %
0	5,990	20,065	22.67	551	15.2	29.9
15	5,450	18,775	21.65	561	15.0	29.0
30	4,540	16,731	19.58	581	14.5	27.1
40	3,930	15,377	18.09	598	14.2	25.6

Weights and Dimensions

Module	Weight (t)	Dimension (m)
Complete unit with GT 7 including gearbox and generator	67	11×3×7
HRSG	~22	~10×3.3×7

FIG. C-59 Data for the GT 7 package for cogeneration application. (Source: Alstom.)



1. Gas Turbine Control Panel
2. Generator Panel
3. Synchronizing and Protection Panel
4. MV Measuring Panel
5. Generator Terminal Box
6. DC System

FIG. C-60 System flowchart: GT 7 gas turbine in cogeneration application. (Source: Alstom.)

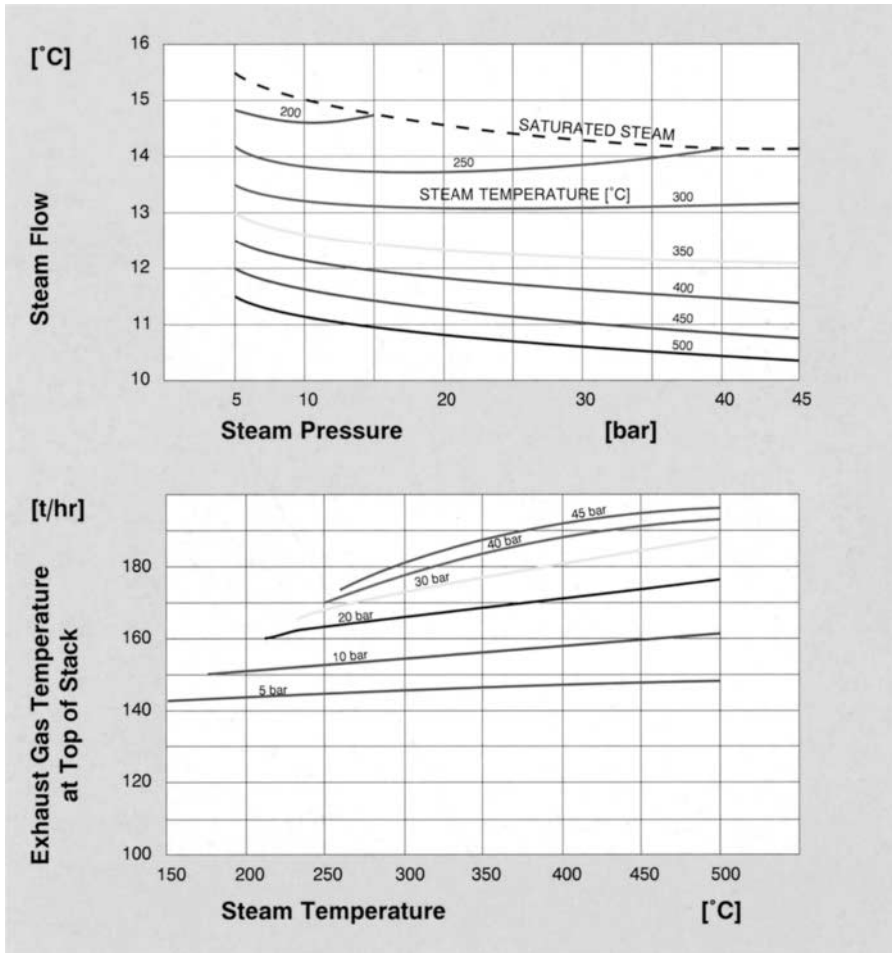


FIG. C-61 Heat recovery steam generator performance curves (GT 7 cogeneration package). (Source: Alstom.)

reinjection application (pushing oil closer to the surface by reinjecting gas in a mixed field).

Where smaller volumes of gas for reinjection purposes are available, a reciprocating design may be used.

Reciprocating*

The most sophisticated design and demanding application among models of reciprocating compressors may be the "laby"™ compressor, which achieves high-pressure ratios without lubrication. This design will be discussed in some detail. An indication of operating ranges for different types of reciprocating compressors may be observed in Fig. C-62.

The "laby" compressor is of sufficient importance to many process applications that the following adapted extracts of a detailed paper are included.

*Source: Sulzer-Burckhardt, Switzerland. Adapted with permission.

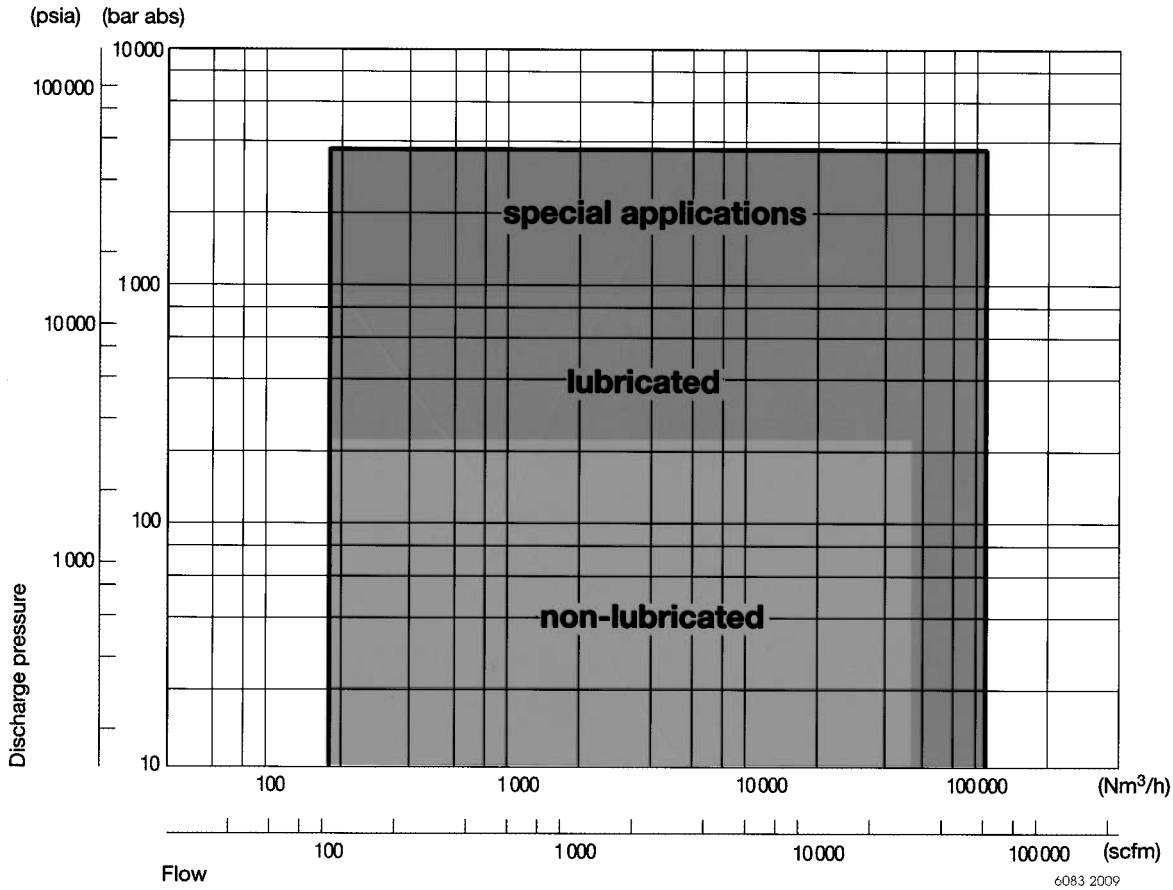


FIG. C-62 Typical performance range of reciprocating process compressors. (Source: Sulzer-Burckhardt.)

Typical applications for "laby" compressors include:

- Petrochemical plants
- Chemical industry
- Natural gas handling
- Polyethylene plants
- Environment protection
- Offshore installations
- Power generation
- Wet oxidation
- Seismic exploration
- Research and development

Typical gases compressed and process conditions:

- Hydrogen
- Hydrocarbon gases

- Hydrogen sulfide
- Hydrogen chloride
- Chlorine
- Highly corrosive gases
- Toxic gases
- Bone-dry gases
- Low-temperature service
- High pressures

Major design and maintenance features

Crankcase and bearings. Rugged compressor frames in cast iron are used for all sizes. This allows an optimized transfer of cylinder forces and rod loads to the frame walls and main bearings. The wall thicknesses are designed to transmit the loads properly and lead to a very stiff frame, thus avoiding uneven elastic deformation between crankcase and cylinders. See Figs. C-63 and C-64. This results in considerable noise reduction and a long life for the crankgear. See also Figs. C-65 through C-68.

Trimetal precision bearings, consisting of a steel shell with lead-bronze layer, babbitt-lined are used. Main bearings and big end bearings are identical and interchangeable. They can be replaced without any manual adjustment. The bearing arrangement ensures that the partition of the two bearing halves is perpendicular to the main load, so that the lubricating film remains unbroken. Oil enters the bearings at points where there is no load and the oil supply bores to the bearings are arranged to maintain an uninterrupted oil flow throughout each complete revolution.

The bearings of heavy-duty process compressors are not considered as wearing parts. The moderate specific load guarantees extremely long bearing life. See Fig. C-69.

Crosshead fastening. The prestressed connection between piston rod and crosshead contains only one connecting element comprising an antifatigue shaft and a screw thread at the piston rod end. The optimized prestress transmits low alternating loads to the thread and can be adjusted entirely without any measuring devices. A simple distance ring is all that is required to properly join the two components. Prestressing is done hydraulically for most compressor sizes requiring no special hydraulic tools. This crosshead fastening element has been developed and is safe. This method by far exceeds the requirements of API 618. See Fig. C-70.

Piston rod runout and crosshead design. A unique procedure is used by this manufacturer to verify the piston rod alignment of horizontal compressors and to reduce the rod runout to a minimum. The system allows the rod runout to be minimized by adjusting the piston rod in relation to the piston by means of a patented eccentric device for the connection of piston and piston rod. It does not need any dismantling of the driving mechanism. This solution is superior to the method recommended in API 618, which advocates crossheads incorporating shimmed shoes. See Fig. C-71.

The easy and quick resetting of the piston rod to its theoretical centerline improves the lifetime of the packing, piston, and guide rings. The need for shoes to



FIG. C-63A Vertical, in-line, reciprocating process compressor, lubricated or oil-free. (Source: Sulzer-Burckhardt.)

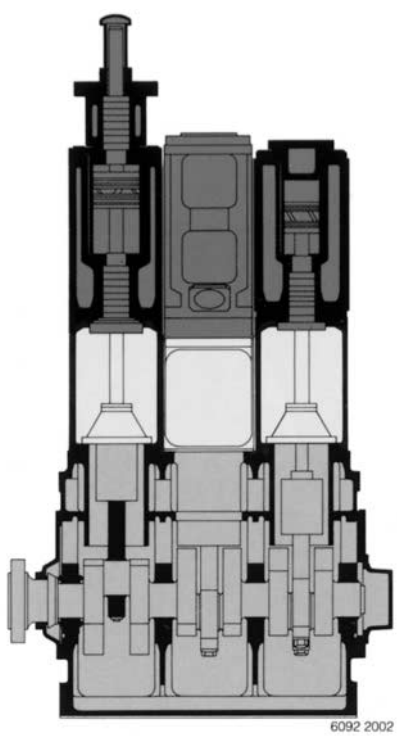


FIG. C-63B Side elevation of Fig. C-63A. (Source: Sulzer-Burckhardt.)

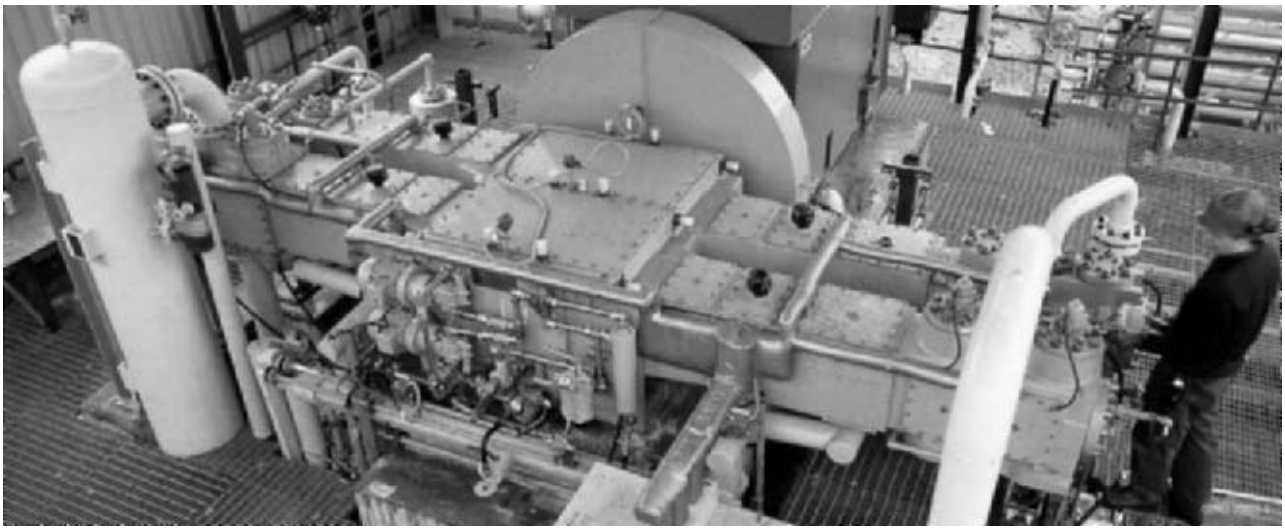


FIG. C-64A Balanced-opposed reciprocating process compressor, lubricated or oil-free. (Source: Sulzer-Burckhardt.)

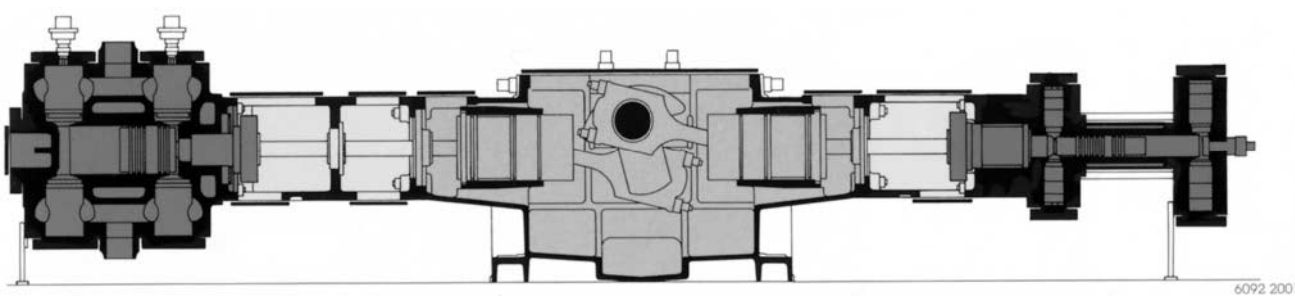


FIG. C-64B Plan view of Fig. C-64A. (Source: Sulzer-Burckhardt.)

be shimmed is rendered obsolete by our patented rod alignment. Therefore we use crossheads with white metal-lined surfaces.

The crosshead is pressure-lubricated. The small end connecting rod bearing runs on a hardened and superfinished full floating crosshead pin. Its specially designed oil feed guarantees a constant unbroken lubricating film during operation.

Piston. A wide variety of piston constructions and materials is available. Depending on the operating condition, we select pistons, sealing rings, wear bands, and piston rods accordingly. See Fig. C-72.

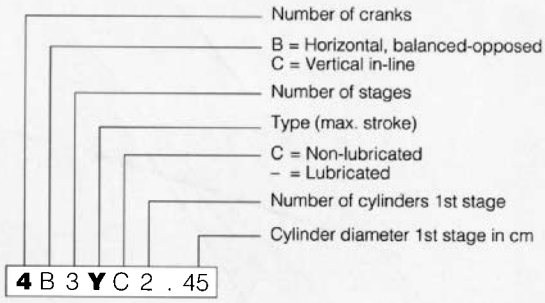
The piston and piston rod connection has a unique and safe locking device. After tightening the piston rod nut to the required prestress, the toothed locking ring can be installed and positively locked in any angular position.

Hollow pistons are continuously self-venting. Nevertheless, they are designed to withstand a differential pressure equal to the maximum cylinder operating pressure.

"Laby" process compressor with benefits for operation and maintenance

Compressor valves. The distillation of many years' experience has guided the design and optimization of compressor plate valves. They contain valve seats and

Key to type designation



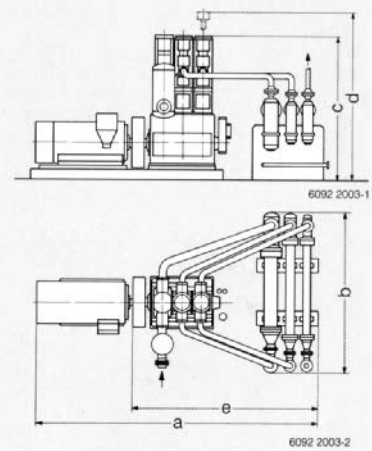
Design data

Type	Max. stroke (mm)	Max. power (kW)	Max. Speed (rpm)
Q	125	300	1 000
Y	160	800	850
S	200	1 600	750
X	270	2 200	600
A	320	3 400	500
C	450	9 500	375
E	560	18 000	300

Main dimensions in millimeters and for estimating purposes only

		a	b	c	d	e
1 crank	**1CQ**	2300	2900	2400	3200	1200
	1CY	3600	4000	3000	4200	1800
	1CS	4200	4600	3400	4600	2000
2 cranks	**2CQ**	3100	2900	2400	3200	1700
	2CY	4800	4000	3000	4200	2900
	2CS	5200	4600	3400	4600	2900
	2CX	7600	6000	3600	5000	5000
	2CA	8100	6500	4300	6000	5200
3 cranks	**3CQ**	4200	2900	2400	3200	2400
	3CY	5900	4000	3000	4200	3900
	3CS	6200	4600	3400	4600	3800
	3CX	8300	6000	3600	5000	5700
	3CA	9000	6500	4300	6000	6100

Vertical



		a	b	c	d	e
2 cranks	**2BQ**	2600	2900	2200	4000	1500
	2BY	3800	4000	2400	6300	1800
	2BS	4800	4600	2600	7500	2600
	2BX	5600	5300	2800	8800	3100
	2BA	6500	5800	3000	10000	3500
	2BC	8300	9000	3600	13000	4800
4 cranks	**4BQ**	3800	2900	2200	4000	2400
	4BY	4700	4000	2400	6300	2700
	4BS	5700	4600	2600	7500	3300
	4BX	6900	5300	2800	8800	4200
	4BA	7700	5800	3000	10000	4700
	4BC	10000	9000	3600	13000	6000
	4BE	12000	11000	4500	15700	8000
6 cranks	**6BS**	6800	4600	2600	7500	4100
	6BX	7800	5300	2800	8800	5000
	6BA	8800	5800	3000	10000	5600
	6BC	11500	9000	3600	13000	7500
	6BE	14000	11000	4500	15700	9700

Horizontal

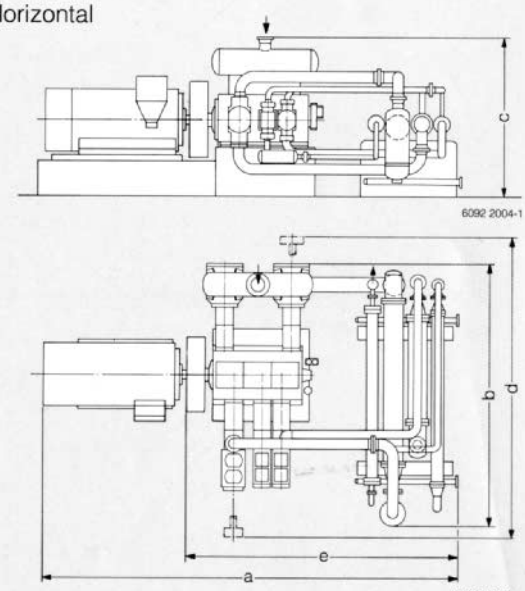


FIG. C-65 Typical reciprocating process compressor design dimensions and corresponding performance range. (Source: Sulzer-Burckhardt.)

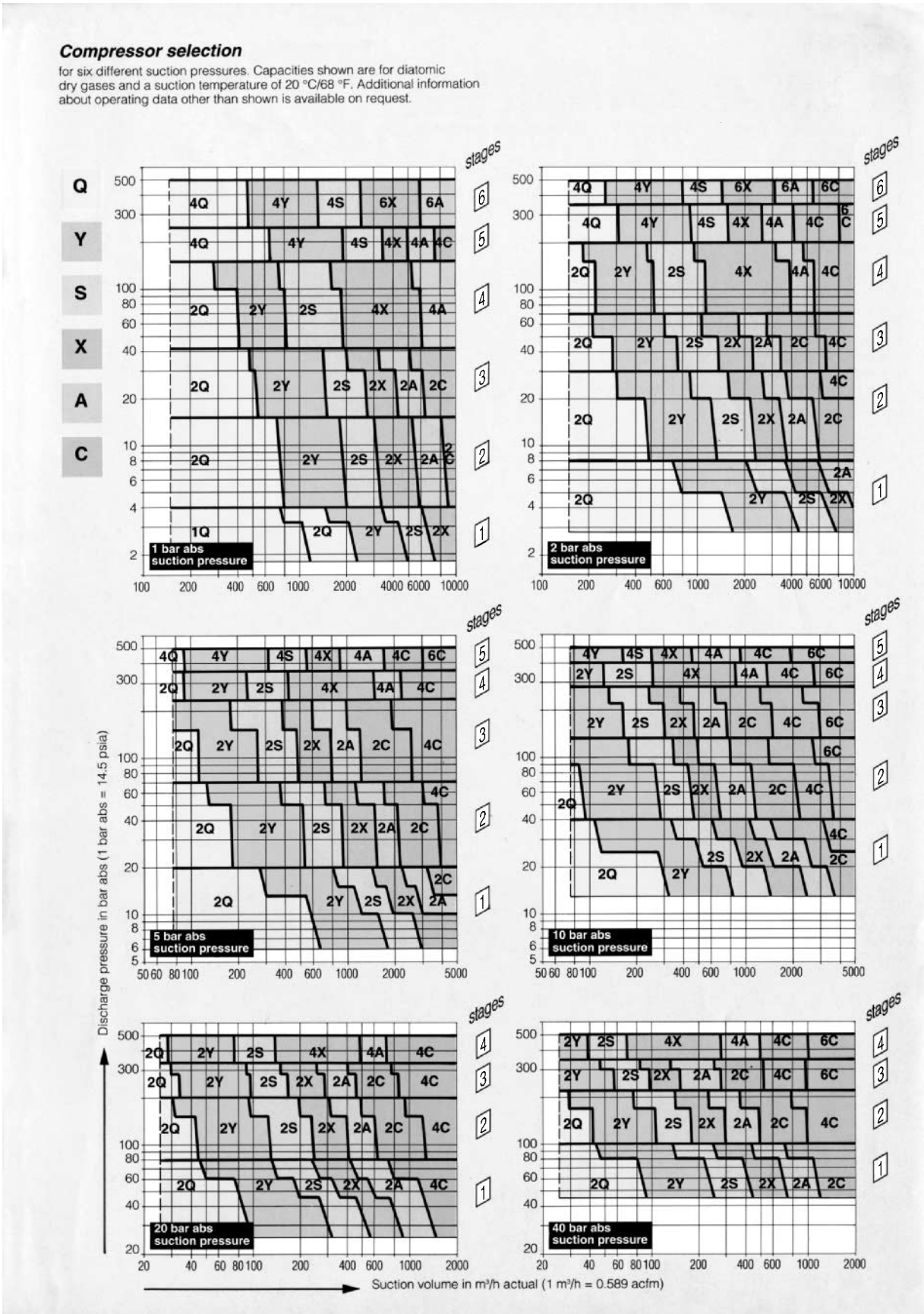


FIG. C-66 Typical compressor selection charts. (Source: Sulzer-Burckhardt.)

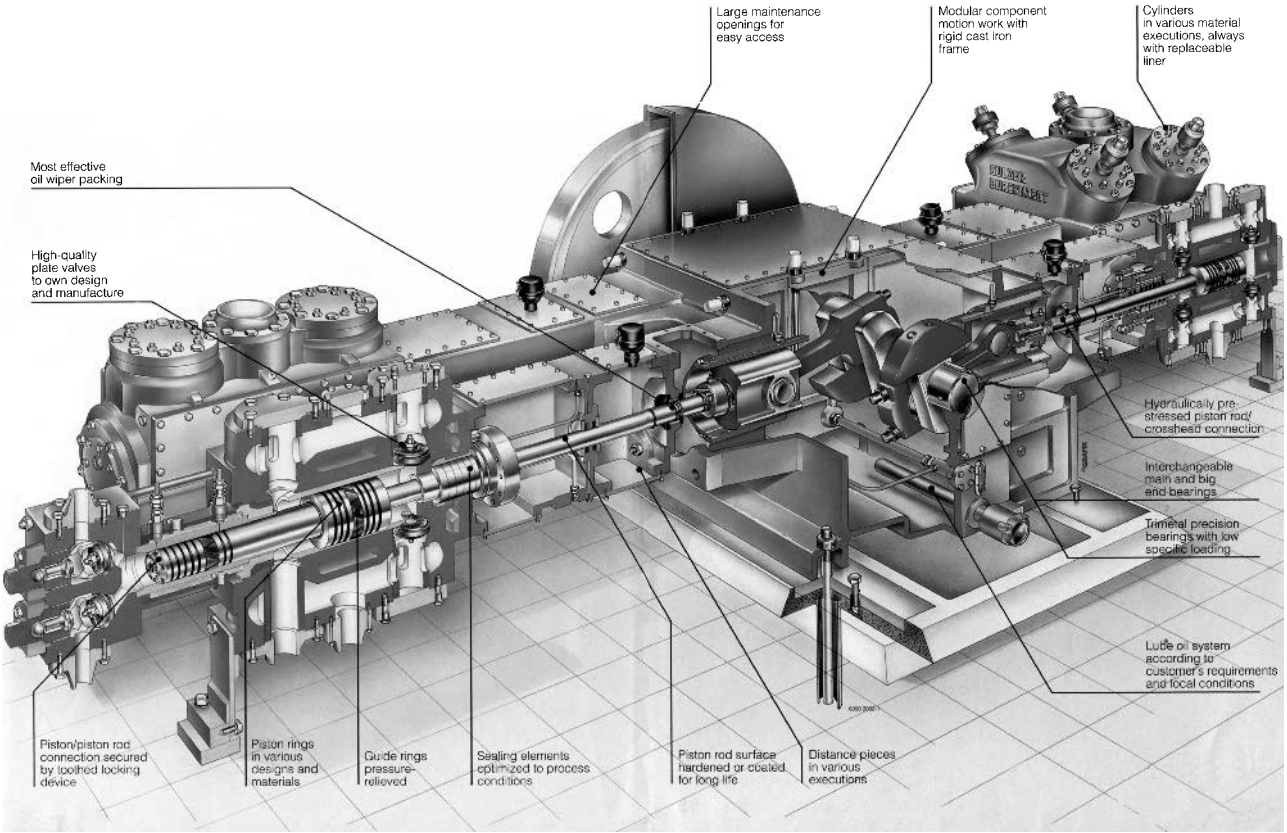


FIG. C-67 Typical features in an API 618 reciprocating compressor. (Source: Sulzer-Burckhardt.)

stroke limiters that are not cast but machined, as well as frictionless guided plates with low lift and good fatigue properties. This OEM has its own compressor valve production facility. This ensures high reliability and availability of the valves and compressors. The valves fulfill the requirements of ISO 8012, API 618, and the European Standard on compressor safety. They are designed to prevent inadvertent wrong assembly into the cylinder. See Fig. C-73.

Suction valves may be equipped with unloading devices for continuous or stepped capacity control.

Special tools. Together with each compressor, the OEM supplies a set of special tools to simplify routine service, like the device to conveniently assemble and install valves for certain sizes of horizontal compressors, or the hydraulic tools, allowing the proper fastening of vital connections. See Fig. C-74.

Piston rod packing. The packing to seal piston rod and cylinder consists of several stacked rings, designed as a cartridge for ease of maintenance. Packing rings are made of PTFE or other newly developed compounds as well as sintered metal. They are subject to stringent quality requirements and feature minimum gas leakage, low friction losses, and minimum wear. See Fig. C-75.

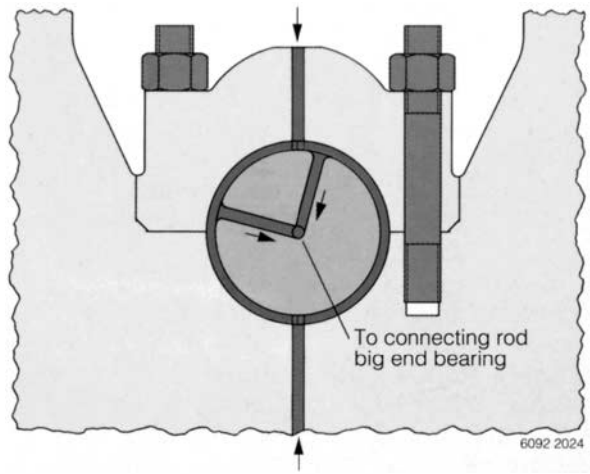




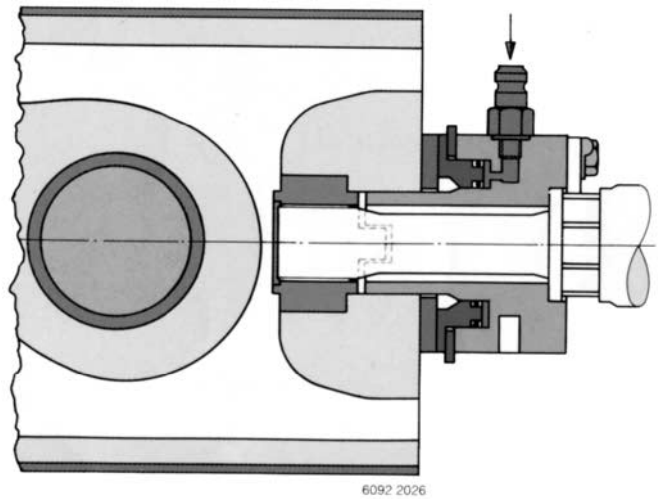
FIG. C-68 Typical crankcase. (Source: Sulzer-Burckhardt.)



FIG. C-69 Typical bearing. (Source: Sulzer-Burckhardt.)

To provide efficient heat removal, direct or indirect cooling is used, depending on the application. Special designs for high-pressure duties or particular process requirements are available.

Piston rod packings are generally well suited to meet the increasing demands imposed by legislation to protect the environment.



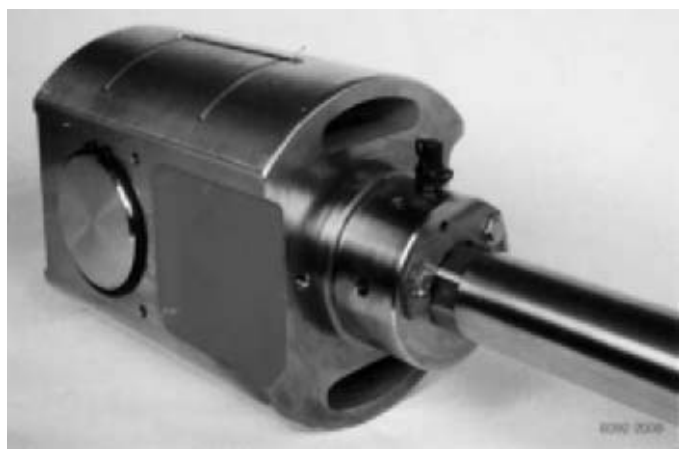
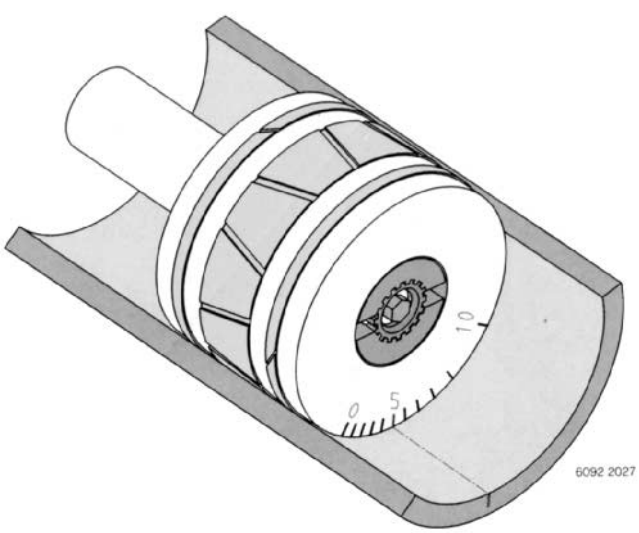


FIG. C-70 Typical crosshead fastening. (Source: Sulzer-Burckhardt.)



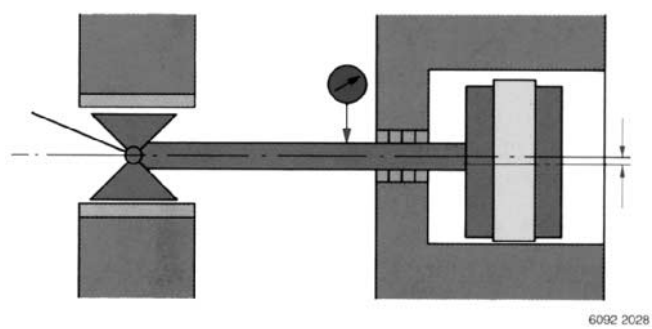


FIG. C-71 Typical piston nod runout and crosshead design. (Source: Sulzer-Burckhardt.)

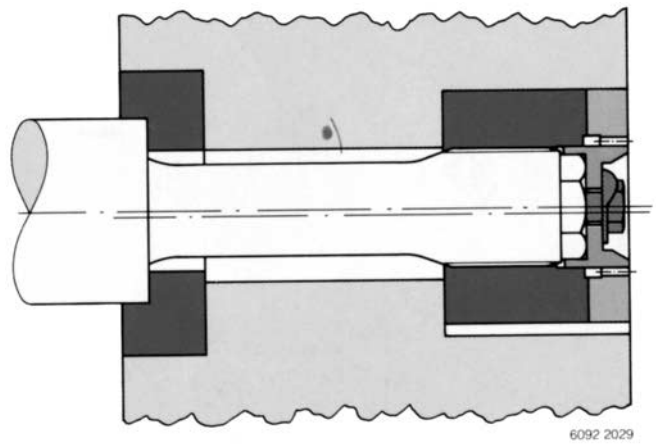




FIG. C-72 Typical piston design. (Source: Sulzer-Burckhardt.)

Oil wiper packing. For oil-free compression no trace of oil may be allowed to creep from the crankcase along the rod into the cylinder. Triple wiper rings—equipped with garter springs—are well known and also used in labyrinth-type compressors for oxygen service. Each of the three rings is provided with an oil wiper lip. The collected oil is led back into the crankcase. See Fig. C-76.

Scraper performance is further improved by providing large communicating passages around the crosshead to avoid crosshead pumping effects.

Next Page